

REQUIREMENT ANALYSIS

Solution Requirements

Date	01 JULY 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	4 Marks

1. Introduction

The **Personalized Networking Assistant (Smart Connect AI)** is an AI-powered web application developed to assist students, professionals, entrepreneurs, and researchers in preparing for networking events. The application uses Artificial Intelligence techniques to analyze event descriptions, generate personalized conversation starters, verify knowledge using the Wikipedia API, save networking sessions, and collect user feedback.

The solution follows a client-server architecture where the Streamlit frontend communicates with the FastAPI backend through REST APIs.

2. Functional Requirements

ID	Requirement	Description
FR-1	User Interface	Interactive Streamlit-based web interface for users.
FR-2	Event Description	Users can enter conference, seminar, workshop, or networking event details.
FR-3	Event Analysis	Analyze event descriptions and extract networking topics.
FR-4	Conversation Starter Generation	Generate AI-powered conversation starters based on the event.
FR-5	Knowledge Assistant	Retrieve verified information using the Wikipedia API.
FR-6	Session History	Save networking sessions locally in JSON format.
FR-7	Feedback Collection	Allow users to submit application feedback.
FR-8	REST API Communication	Enable communication between frontend and backend.
FR-9	Error Handling	Display meaningful error messages during failures.
FR-10	Responsive UI	Provide a clean, attractive, and responsive interface.

3. Non-Functional Requirements

ID	Requirement	Description
NFR-1	Performance	Generate conversation starters quickly.
NFR-2	Reliability	Provide accurate responses and stable execution.
NFR-3	Availability	Application should be available whenever the server is running.
NFR-4	Security	Protect stored session data and feedback.
NFR-5	Scalability	Easily extend the application with additional AI features.
NFR-6	Maintainability	Modular Python code for future development.
NFR-7	Usability	Simple interface that requires minimal training.
NFR-8	Portability	Compatible with Windows, Linux, and macOS.

4. Software Requirements

Component	Technology
Programming Language	Python 3.x
Frontend	Streamlit
Backend	FastAPI
API Communication	Requests Library
Knowledge Source	Wikipedia API
Data Storage	JSON Files
IDE	Visual Studio Code
Version Control	Git & GitHub
Web Server	Uvicorn
Operating System	Windows 10/11

5. Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher
RAM	Minimum 4 GB (8 GB Recommended)
Storage	2 GB Free Disk Space
Internet Connection	Required for Wikipedia API access
Browser	Chrome, Edge, Firefox

6. Input Requirements

The system accepts the following inputs:

- User Name
 - Profession
 - Networking Goal
 - Event Description
 - Knowledge Topic
 - User Feedback
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7. Output Requirements

The application produces:

- AI-generated conversation starters
 - Verified information from Wikipedia
 - Saved networking session confirmation
 - Feedback submission confirmation
 - Success and error notifications
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8. System Constraints

- Internet connection is required for knowledge verification.
- FastAPI backend must be running before launching Streamlit.
- Python dependencies must be installed.
- JSON files are used for local data storage.

9. Assumptions

- Users have basic knowledge of using a web application.
 - The backend server is accessible.
 - Wikipedia API is available.
 - Required Python packages are installed.
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10. Expected Benefits

The proposed system provides:

- Intelligent networking assistance
 - Faster event preparation
 - AI-powered conversation suggestions
 - Reliable topic verification
 - Improved communication skills
 - Networking history management
 - User-friendly interface
 - Enhanced professional networking experience
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11. Conclusion

The **Personalized Networking Assistant (Smart Connect AI)** fulfills all functional and non-functional requirements for an AI-powered networking support platform. By integrating **FastAPI**, **Streamlit**, **Wikipedia API**, and **JSON-based storage**, the system provides a reliable, scalable, and user-friendly solution that helps users prepare confidently for conferences, seminars, workshops, career fairs, and other professional networking events.